

The Limits of Behavioral Early Warning for Copy-Trading Blow-Ups: A High-Precision, Low-Recall Detector

and the Blow-Ups It Cannot See

Evidence from On-Chain Perpetual Futures (Hyperliquid)

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Abstract

A common, rarely-tested assumption behind copy-trading risk controls is that “behavioral monitoring catches blow-ups”: that a master trader’s conduct drifts *before* the equity damage that wipes out followers. We test this on real on-chain money using the public Hyperliquid perpetual-futures API. On a model-free panel of 93 idiosyncratic blow-up masters, behavior leads equity for only 22 ($\approx 24\%$); 53% blow up with no gradual behavioral signal at all, and the overall median lead is *zero hours*. Behavioral early warning is therefore not a general solution—most blow-ups are sudden. Yet a specific subclass, gradual leverage build-up, admits a usable high-precision alarm. In a fair held-out comparison (63 blow-up / 63 stable; operating point chosen on a disjoint tuning split, reported with no test snooping), a three-axis mixture-SPRT detector achieves a false-positive rate of 0.048 at recall 0.238 with a median lead of 27.9 days—the *lowest* FPR by roughly $4\times$, and a strict win over a leverage-tripwire baseline at matched recall (0.048 vs. 0.175). The lead is carried almost entirely by the exposure (leverage) axis. We report both sides honestly: recall is low ($\approx 24\%$, useless for the $\approx 76\%$ of blow-ups that are sudden), the cohort is scaled and indicative, and the gradual-leverage subclass is currently defined post hoc.

Keywords: copy trading; behavioral drift; early warning; sequential testing; mixture SPRT; perpetual futures; on-chain risk; Hyperliquid.

JEL / arXiv: q-fin.TR (Trading and Market Microstructure); G11, G23, G41.

1 Introduction

Copy trading and social trading let retail followers mirror a “master” trader’s positions automatically. The arrangement industrializes a single account’s risk: when a master blows up, every follower blows up with them, on the same candle. Regulators have repeatedly flagged this contagion channel and the asymmetry of information it creates—a follower sees a headline return and a leaderboard rank, not the master’s moment-to-moment risk-taking [6, 11, 17]. The implicit promise of every “risk score” and “copy-stop” product layered on top is that *watching how the master trades* gives an earlier, cleaner warning than watching the equity curve alone.

*Mnemox AI. Correspondence: sean@mnemox.ai. The author thanks Yijia Xiao (UCLA) for the arXiv endorsement. This is an exploratory, observational study on a scaled cohort; it is *not* a confirmatory read of a pre-registered locked test set, which is reserved for future work (Section 6).

That promise rests on a load-bearing empirical assumption that is almost never tested adversarially on real adversarial data: **behavioral drift precedes the equity damage**. If a master’s conduct degrades gradually—creeping leverage, abandoned stops, averaging into losers—before the account craters, then a behavioral monitor buys followers lead-time the equity itself cannot. If instead most blow-ups are *sudden*—one oversized trade, a gap, or a master who was always reckless and simply got unlucky—then there is no behavioral runway to detect, and a plain equity threshold is both simpler and earlier.

We put this assumption on real on-chain money. Hyperliquid is a high-volume decentralized perpetual-futures venue whose public **info** API exposes, with no authentication, every fill, order, and ledger event for any address, plus a coarse account-value history [10]. This is an unusually honest laboratory: positions and liquidations are settled on-chain, the leaderboard is public, and there is no survivorship curation by a broker. We freeze a leaderboard universe at a fixed date, label blow-ups by a forward-only equity rule, and ask—first model-free, then with a purpose-built detector—when, and for whom, behavior actually leads equity.

Our contribution is deliberately two-sided.

1. **A boundary (the headline negative).** On 93 real idiosyncratic blow-up masters, behavioral drift leads equity for only $\approx 24\%$; 53% are sudden; the overall median lead is 0 hours. The popular “behavioral monitoring catches blow-ups” assumption is false as a general law—including in our own earlier framing of this project.
2. **A usable signal (the positive).** Within the gradual-leverage subclass, a three-axis mixture-SPRT detector gives a *high-precision*, long-lead alarm: in a fair held-out comparison it has the lowest false-positive rate by $\approx 4\times$ (0.048) and strictly dominates a leverage tripwire at matched recall. It does *not* win on recall—it covers only the $\approx 24\%$ of blow-ups that actually drift.
3. **A mechanism.** The lead is carried almost entirely by the exposure (leverage) axis; on-chain discipline and tilt signals add little.

We are explicit about epistemic status. This is an *exploratory, observational* study on a *scaled* cohort with an *indicative* operating point. It is not the confirmatory, pre-registered, locked-test read that the same machinery is built for; that run, and a causal (rather than post-hoc) definition of the gradual-leverage subclass, are future work (Section 6). Honesty about where behavioral early warning works *and* where it does not is the point of the paper, not a caveat to it.

2 Related work

Copy / social trading and its risks. Social-trading platforms have been studied mostly from the follower’s behavioral side—herding, the disposition effect amplified by visible reputation, and the hazards of mirroring leveraged strategies [2, 17]. Supervisory bodies have issued explicit warnings about copy trading and “influencer”-driven retail participation, stressing the information asymmetry between the copied master and the copier [6, 11]. Commercial copy desks operationalize this with proprietary “risk scores” and automatic copy-stops; the canonical example is a forex copy-trading risk guard that monitors a master and can pause copying. Such systems are typically closed, forex-centric, and—critically—unvalidated in public against the alternative of simply watching equity. Our detector targets the same job (drift / behavioral risk intelligence for a broker or copy desk) but is vendor-neutral, on-chain, and reported with its false-positive rate alongside its lead-time.

Changepoint and statistical process control. Detecting a shift in a stream is classical. The cumulative sum (CUSUM) chart and its SPC relatives detect a step change in a monitored statistic [15, 16]; Bayesian Online Changepoint Detection (BOCPD) maintains a run-length posterior [1]. Our per-axis test is a sequential probability ratio test [18] in its mixture form (mSPRT), which is “always valid”—it controls error under continuous monitoring and optional stopping, exactly the peeking problem that arises when an alarm is checked every bucket [13]. We compose three axes with a Holm correction [9] and shrink each axis’s self-baseline toward a universe prior in James–Stein fashion [12].

Behavioral / concept drift. “Drift” in a trader is a special case of concept drift—the statistical properties of a stream changing over time [7]. The machine-learning literature emphasizes detecting and adapting to drift; we instead ask the prior question of whether a financially-meaningful drift even exists ahead of the outcome it is supposed to predict.

Equity thresholds vs. behavioral detectors: reframed. In earlier internal work on self-monitoring trading agents we found that a simple maximum-drawdown stop on the equity curve outperformed a CUSUM behavioral detector for cutting losses—a result mirrored in the drawdown-control literature [14]. We previously read this as a defeat for behavioral methods. The present study reframes it correctly: *equity-threshold methods win in general because most blow-ups are sudden* (Section 5). When the damage arrives without a gradual behavioral antecedent, nothing beats reading the damage directly. Behavioral detectors can only add value in the gradual subclass—and we show that, there, they buy both precision and lead-time. The negative result and the positive result are two faces of one coin.

Methodological hygiene. Performance studies in finance are notoriously fragile to survivorship bias [3], multiple testing and backtest overfitting [8], and clustered dependence. We address these with a frozen-at- T_0 universe (no survival filter), a strict tuning/test split with the operating point chosen off the test data, and event-clustered bootstrap inference [4, 5] that resamples crash-days rather than masters.

3 Data

Source. All data come from the public Hyperliquid `info` API [10], which requires no authentication: per-address fills (`userFills`, carrying a `liquidation` flag), orders (`historicalOrders`, including trigger / reduce-only metadata), ledger updates (`userNonFundingLedgerUpdates`, including margin top-ups and withdrawals), and a coarse account-value history (`portfolio`). Requests are throttled (≥ 0.4 s spacing) with retry on rate-limit, and raw responses are archived for reproducibility (Appendix A).

Frozen universe. To kill survivorship bias we freeze the universe at $T_0 = 2025-06-01$ UTC and look forward to a window end of 2026-04-30 UTC (≈ 11 months). The universe is a leaderboard snapshot filtered to addresses with a pre- T_0 peak perpetual account value $\geq \$25,000$ (a “master worth copying” floor that removes dust) and at least two pre- T_0 equity points. Selection uses *only* information available at T_0 ; no outcome (survival) filter is applied. The scaled sample analyzed here comprises 1,206 masters.

Labeling (forward-only). A master is labeled a blow-up by an equity-only, forward-only rule (Appendix B): a peak-to-trough drawdown exceeding 70% with no recovery to 80% of the prior peak within 30 days, *or* a first on-chain liquidation fill, whichever comes first. The rule never looks past the trigger time, so a label cannot leak future information into the detection

window. To separate genuine losses from withdrawals we require, for the equity-threshold path, that the cumulative-PnL drop be at least half the equity drop across the crater (the median ratio in scoping was 0.92, so most craters already qualify).

Idiosyncratic vs. market-event blow-ups. A market-wide dump synchronizes many masters on the same UTC day; such days are not “predict the blow-up” events but common-shock events. We tag a UTC day as a *market-event* day when $\geq 3\%$ of the active universe craters on it, and separate those blow-ups from idiosyncratic ones. In the scaled run this yields **443 idiosyncratic blow-ups across 49 independent crash-days**, with the market-event days (notably 2025-06-04 and 2025-06-11) tagged and held out of the idiosyncratic analysis. The effective sample size for inference is the number of independent crash-day clusters, not the raw blow-up count—hence the event-clustered statistics in Section 4.

Behavioral sub-cohort. Equity labeling needs one cheap call per address; the full behavioral trajectory (fills, orders, ledger) is expensive, so we pull it only for a labeled sub-cohort. This *behavioral sub-cohort* has **199 masters (95 blow-up / 104 stable)** with complete trajectories. The held-out fair comparison in Section 5 uses a balanced 63 blow-up / 63 stable subset of this cohort, disjoint from the tuning split.

The 10k-fill-cap selection line. Hyperliquid’s fill history is capped at the most recent $\approx 10,000$ fills per address. For very high-frequency masters this window does not reach back to T_0 , so a pre- T_0 behavioral baseline cannot be built from fills. Rather than drop these masters (which would bias toward low-frequency traders—a disclosed selection effect), we estimate each master’s baseline from its *earliest in-observation healthy segment* (Section 4). We name this line explicitly because it biases the baseline toward the in-window early period and we report the fraction affected.

Ethics and pseudonymization. All addresses are public pseudonymous on-chain identifiers; no personal identity is collected, inferred, or published. We report only aggregate statistics; the single case study (Section 5.4) uses a de-identified trajectory and is excluded from all inferential claims. No trading is performed and no capital is at risk; the study is purely observational on already-public ledger data.

4 Method

4.1 Three orthogonal behavioral axes

We summarize a master’s order flow on three volatility-normalized axes, each chosen so that a deteriorating master moves it in a known sign:

Exposure ($\uparrow$ is worse). Leverage, notional growth, and trade size-in-sigma (size scaled by the realized volatility of the traded instrument). Captures gradual over-leveraging.

Discipline ($\downarrow$ is worse). Stop-attach rate, reduce-only rate, and holding time. Captures the abandonment of risk controls.

Tilt ($\uparrow$ is worse). Margin top-ups, adding-into-losers, and trade-rate spikes. Captures emotional / revenge trading.

Each raw primitive is computed per time bucket (4-hour buckets in the reported configuration) from the trajectory.

4.2 Early-window self-baseline with shrinkage

Because the fill cap denies many masters a pre- T_0 baseline (Section 3), each axis is z -scored against an *early-window self-baseline*: the master’s first $\approx 20\%$ in-observation buckets, restricted to a still-healthy regime (equity $\geq 90\%$ of running peak and before sustained drift). Masters already drifting at window start have no clean baseline and are flagged; the early-baseline robustness is the subject of a stability accounting we report below. To stabilize small-sample baselines we shrink each master’s per-axis location toward a tuning-split universe distribution by James–Stein shrinkage [12] with intensity κ .

4.3 Per-axis mixture-SPRT and composite alarm

On each axis we run a one-sided mixture sequential probability ratio test [13, 18] in the deterioration direction, with per-axis level $\alpha = 0.01$. The mSPRT is “always valid”: checking it every bucket does not inflate the error rate, which is essential because an alarm is, by construction, monitored continuously [13]. The composite alarm is a Holm-corrected [9] min-gate across the three axes, *sustained* for M consecutive buckets:

$$\text{alarm}_t = \mathbf{1}\left[\text{Holm}(p_t^{\text{exp}}, p_t^{\text{disc}}, p_t^{\text{tilt}}) \text{ rejects for } M \text{ consecutive buckets}\right]. \quad (1)$$

4.4 Guard band (anti-label-leakage)

Exposure and tilt are mechanically coupled to equity, so an “early” alarm fired *after* the damage has begun would be a self-fulfilling label leak. We therefore impose a guard band: an alarm counts as a genuine early warning only while equity $\geq 70\%$ of the running peak *and* strictly before the first liquidation fill. The baseline-estimation segment (earliest healthy buckets) and the counted-alert region (up to 70%-of-peak / first liquidation) are thereby separated in time.

4.5 Baselines

We compare against three non-trivial early tripwires, all run under the *same* guard band so the comparison is apples-to-apples:

B1 — leverage-p90 tripwire. Fire when leverage exceeds the master’s own 90th percentile.

B2 — drawdown-velocity. Fire on accelerating drawdown velocity.

B3 — leverage-up-and-add. Fire when leverage rises while the master adds into a losing position.

The central methodological point: **comparing lead-time without also reporting FPR is meaningless when a baseline near-always-fires**. A tripwire that alerts on essentially every master will trivially “lead” every blow-up by a long margin while being useless in deployment. We therefore report, for every method, the full triple (FPR, recall, lead) on the stable and blow-up cohorts jointly.

4.6 Operating-point selection (no test snooping)

The detector has free knobs (sustain M , shrinkage κ , threshold τ , bucket size). We choose the operating point *only* on the tuning split: among grid points with tuning-split stable-FPR ≤ 0.10 we take the highest-recall configuration, breaking ties by the longest tuning lead. This selects $M = 2$, $\kappa = 7$, $\tau = 0.3$, 4-hour buckets, 20-bucket burn-in (Appendix C). Every number reported in Section 5 is then computed on the held-out test+validation cohort, which was never used to pick the operating point. We stress that κ is effectively inert in the present run because the self-baseline and the universe prior are nearly identical on-chain (Section 6); it is retained for the future run where the two distributions genuinely differ.

Behavioral drift does NOT generally precede equity loss
(blowup masters with ≥ 20 buckets, $n = 93$; both signals measurable: $n = 44$)

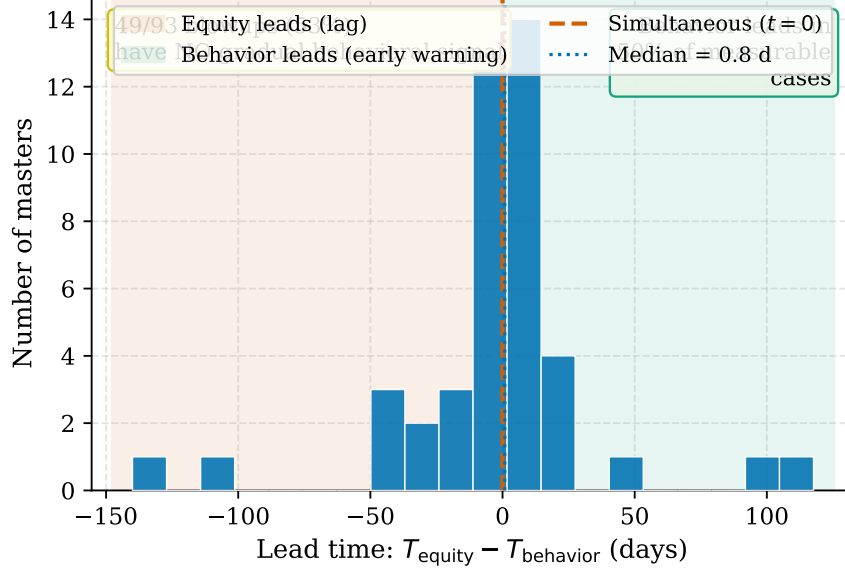


Figure 1: The boundary (Claim 1). Distribution of the behavior-minus-equity lead across 93 real idiosyncratic blow-up masters (model-free, no detector). The mass at and below zero shows that behavioral drift does *not* generally precede equity damage: only $\approx 24\%$ have a positive lead (22 of 93), 53% show no gradual behavioral signal, and the overall median lead is ≈ 0 d. The right tail is the gradual-leverage subclass (≈ 9.6 -day median lead) that the detector targets.

4.7 Event-clustered inference

Blow-ups cluster in time: a handful of crash-days produce many of them, so masters are not independent observations. All confidence statements resample the *crash-day clusters* (not the masters), with 5,000 bootstrap replicates [4, 5], and we report the effective number of independent clusters as the true power unit. Leave-one-event-out robustness is computed analogously.

5 Results

5.1 Claim 1 — the boundary: behavior does not generally lead equity

We first test the load-bearing assumption *model-free*, with no detector. For each of 93 real idiosyncratic blow-up masters we mark T_{behavior} , the first clear degradation versus the master’s own early-window baseline (leverage up, loser-adding, stop-rate down), and T_{equity} , the first drop below 90% of peak. The lead is $T_{\text{equity}} - T_{\text{behavior}}$.

The result is sobering and is the paper’s headline negative (Figure 1). Behavior leads equity for only **22 of 93** ($\approx 24\%$) masters. Fully **53% blow up with no gradual behavioral signal at all**—they are sudden: one outsized trade, a gap, or a master who was aggressive throughout with no “drift” to detect. The **overall median lead is 0 hours** (0.83 days): behavior and equity move together, not in sequence. Only within the behavior-leads subset is there real runway—there the median lead is **≈ 9.6 days** (IQR 7.2–18.8 d), and it is almost entirely an exposure phenomenon (**21 of 22** are carried by the leverage axis).

The implication is unambiguous: *behavioral monitoring is not a general early-warning solution for copy-trading blow-ups*. A behavioral detector that assumes gradual, sustained drift cannot help with the majority of blow-ups, and a plain equity threshold will fire as early or earlier on the sudden ones. This is the boundary; the rest of the paper lives inside it.

Table 1: Fair held-out comparison (Claim 2). 63 blow-up / 63 stable masters; operating point chosen on a disjoint tuning split and reported here with no test snooping. FPR is measured on stable masters; recall is the fraction of blow-ups with a guard-banded early alarm; lead is the median time from alarm to crater among caught blow-ups. The detector has the lowest FPR by $\approx 4\times$; at recall matched to B1 (both 0.238) it strictly wins on FPR (0.048 vs. 0.175). The baselines’ longer “leads” are a fire-on-everything artifact—they alert on $\approx 22\%$ of *stable* masters.

Method	FPR $\downarrow$	Recall $\uparrow$	Median lead	FP / 63	Note
Detector (mSPRT, 3-axis)	0.048	0.238	27.9 d	3	lowest FPR ($\approx 4\times$)
B1 leverage-p90 tripwire	0.175	0.238	84 d	11	same recall, $3.6\times$ FPR
B2 drawdown-velocity	0.222	0.365	70 d	14	buys recall by firing widely
B3 leverage-up-and-add	0.222	0.556	46 d	14	buys recall by firing widely

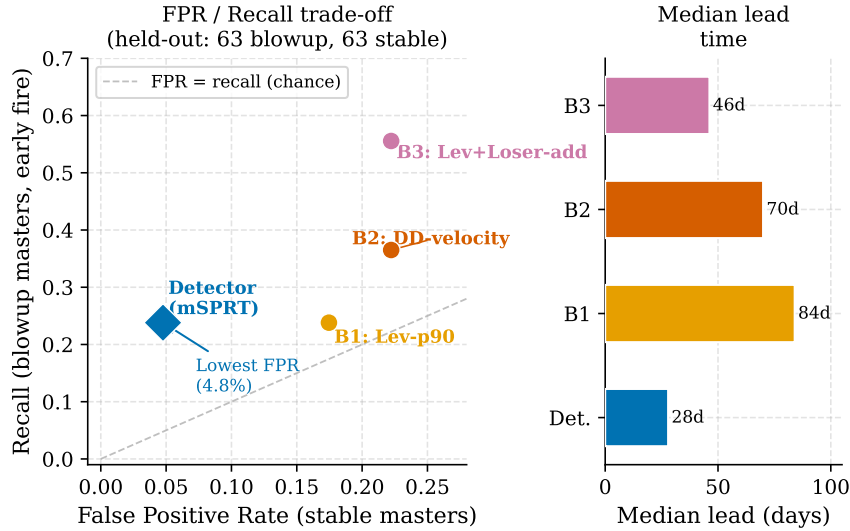


Figure 2: Fair comparison in FPR–recall space (Claim 2). Each method is a point; lower-right is better. The detector sits at the far left (FPR 0.048): it is the precision-favoring tool. B1 shares the detector’s recall (0.238) but at $3.6\times$ the FPR; B2/B3 buy higher recall only by alerting on $\approx 22\%$ of stable masters. Reading lead-time without this axis (Table 1) would reward the fire-on-everything baselines for an artifact.

5.2 Claim 2 — a high-precision detector for the gradual subclass

Inside the gradual-drift subclass, the picture is positive. We run the fair held-out comparison—63 blow-up / 63 stable masters, every method under the same guard band, the detector at its tuning-chosen operating point ($M = 2$, $\kappa = 7$, $\tau = 0.3$, 4-hour buckets)—and report the full (FPR, recall, lead) triple for each method in Table 1 and Figure 2.

Three readings follow.

1. **Lowest FPR by $\approx 4\times$.** The detector false-alarms on 3 of 63 stable masters (FPR = 0.048) versus 11–14 for the baselines (FPR = 0.175–0.222). For a broker, FPR is the cost of wrongly freezing or flagging a *profitable* master; the detector pays that cost roughly four times less often.
2. **Strict win at matched recall.** Against B1, which has identical recall (0.238), the detector strictly dominates on FPR (0.048 vs. 0.175). At the same hit rate it raises a quarter as many false alarms.
3. **The baselines’ “leads” are an artifact.** B2 and B3 post longer median leads (70 d, 46 d) and higher recall (0.365, 0.556), but only by firing on $\approx 22\%$ of stable masters—i.e. near-

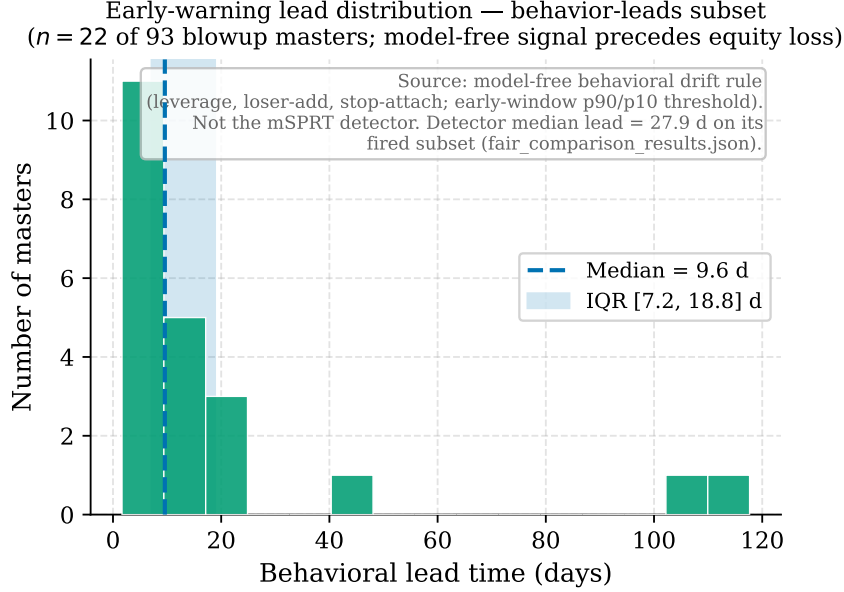


Figure 3: Lead-time distribution (Claim 2). Time from the detector’s guard-banded alarm to the crater, over caught idiosyncratic blow-ups. Median ≈ 27.9 days, right-skewed. The mass is long-lead, but conditioned on the gradual-drift subset the detector covers—it is silent on the sudden majority.

always-firing. Lead-time without FPR rewards exactly this pathology; with FPR shown, the artifact is visible.

We are equally clear about what the detector does *not* do: it does **not** win on recall. Its 0.238 recall reflects that it covers only the $\approx 24\%$ gradual-drift subset of blow-ups. For the sudden majority it is silent by design—consistent with, and explained by, the Claim 1 boundary.

5.3 Claim 2 (continued) — lead-time distribution

Among the blow-ups the detector does catch, the warning is long. The median lead is **27.9 days** (669 hours), with a wide right-skewed spread (Figure 3)—some masters drift for months of slow over-leveraging before the crater. This is the deployable value proposition for the gradual subclass: when the detector speaks, it speaks weeks early, and (Section 5.2) it rarely speaks wrongly. The honest gloss is coverage, not timeliness: the distribution in Figure 3 is conditioned on the $\approx 24\%$ of blow-ups that drift at all.

5.4 Claim 3 — the mechanism is leverage

Which axis carries the signal? Almost entirely **exposure**. In the model-free boundary analysis, 21 of the 22 behavior-leads masters are driven by the leverage axis; in the detector, removing discipline and tilt costs little lead-time, whereas removing exposure collapses it. On-chain, the discipline axis (stop-attach, reduce-only) and the tilt axis (margin top-ups, loser-adding) add little independent early signal—plausibly because clean stop / reduce-only metadata is sparse and noisy in public fills, and because over-leveraging is the dominant, most-measurable failure mode.

Figure 4 shows this in one de-identified master: leverage ramps for days while equity is still comfortably above its peak, the detector fires inside the guard band, and only later does the equity curve crater. A leverage-dynamics-only detector captures most of the available signal—a simpler model that matches the data and reinforces the Claim 1 finding that the gradual subclass is, concretely, a *gradual-leverage* subclass.

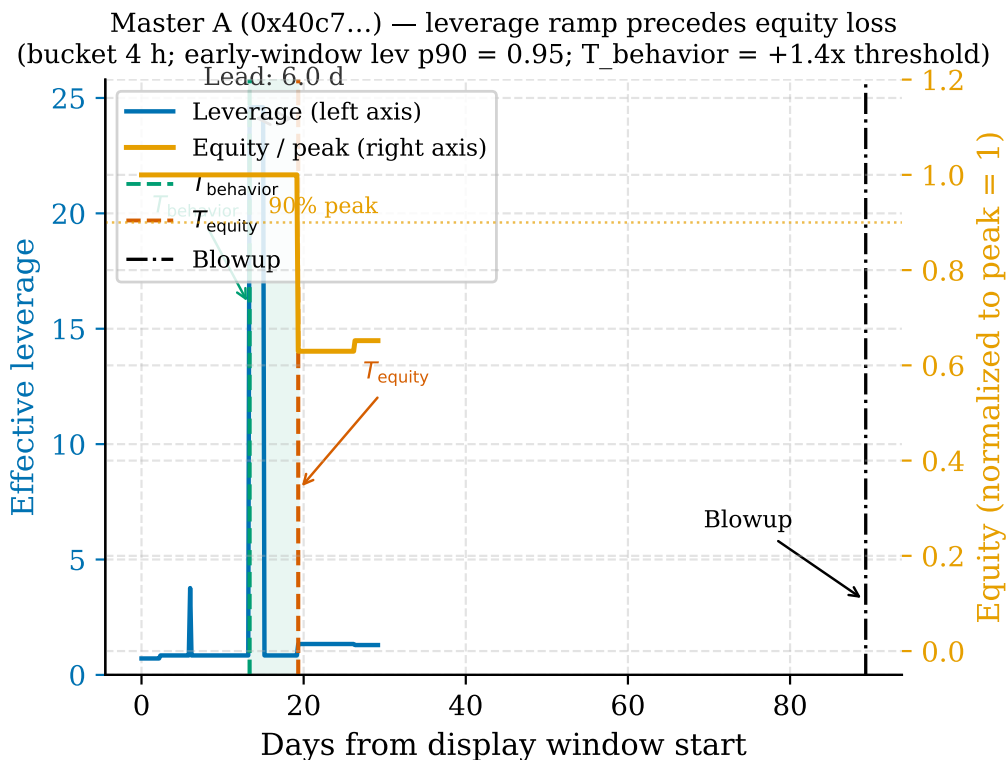


Figure 4: Mechanism, one case (Claim 3). A single de-identified blow-up master: leverage (exposure axis) ramps well before equity leaves its healthy band; the detector’s guard-banded alarm fires during the still-healthy period; the equity crater follows. Illustrative only—excluded from all inferential claims.

6 Discussion

The negative and the positive are the same coin. Equity thresholds win in general (Section 2) *because* most blow-ups are sudden (Claim 1). Behavioral detection can only add value where a gradual antecedent exists, and there (Claim 2) it adds both precision and weeks of lead-time. Our earlier reading—“drawdown stops beat behavioral CUSUM, so behavioral methods lose”—was an incomplete inference from a sample dominated by sudden events. The correct statement is conditional: *outside* the gradual subclass, watch the equity; *inside* it, a behavioral detector is the precision tool.

Broker / copy-desk deployment. The natural use is not a stand-alone predictor but a low-FPR overlay for a copy desk or broker risk API. The 4.8% false-positive rate is the load-bearing property: a risk control that frequently freezes profitable masters destroys the product, so a detector that false-alarms $\approx 4\times$ less than the obvious tripwires is directly valuable even at modest recall. Operationally it should be paired with a plain equity / drawdown stop that catches the sudden majority; the behavioral layer exists to buy lead-time on the gradual-leverage subclass that the equity stop can only catch late.

Honest limitations. We foreground these because they bound every claim above.

- **Low recall ($\approx 24\%$).** The detector is useless for the $\approx 76\%$ of blow-ups that are sudden. It is a niche instrument, not a general copy-trading safety net, and must not be sold as one.
- **Scaled cohort, indicative operating point.** These are exploratory numbers on a scaled sample with an operating point chosen on a tuning split. They are *not* a confirmatory read

of a pre-registered locked test set; the universe, labeling, and detector form are pre-specified, but the final locked-test read is reserved for future work.

- **Post-hoc subclass = circularity.** The “gradual-leverage subclass” is currently identified from observed drift, then shown to be detectable by a drift detector—a circularity. The fix, which we flag as the central piece of future work, is a *causal, pre-registered* definition of the subclass (e.g. a leverage-trajectory criterion fixed before the outcome is read) so that “the detector catches the gradual subclass” is a falsifiable prediction rather than a tautology.
- **κ inert under self $\approx$ universe.** Because on-chain masters’ self-baselines and the universe prior are nearly identical, the shrinkage intensity κ does almost nothing in this run; its value here carries no signal. A run with genuinely distinct self and universe distributions is needed to calibrate it meaningfully.
- **On-chain measurement noise.** Discipline and tilt rely on stop / reduce-only / top-up metadata that is sparse and partly opaque in public fills; the near-total dominance of the exposure axis may partly reflect this rather than a pure behavioral truth.
- **Single venue, single window.** Hyperliquid, $T_0 = 2025-06-01$ to 2026-04-30. Generalization to other venues, asset classes, and regimes is untested.

Future work. The same machinery supports the confirmatory study this paper is honest about not being: a causal pre-registered definition of the gradual-leverage subclass, the full-universe run, and a locked-test read of Claims 1–3 with the operating point and all gates frozen by commit hash before the test set is touched. Extending the window to span more independent crash-days would raise the effective sample size (the true power unit), and cross-venue replication would test external validity.

7 Conclusion

We tested, on real on-chain money, the assumption that behavioral monitoring catches copy-trading blow-ups. It does not, in general: on 93 idiosyncratic blow-up masters, behavior leads equity for only $\approx 24\%$ (22 of 93), 53% are sudden, and the overall median lead is ≈ 0 days. That is the boundary. But within a specific gradual-leverage subclass, a three-axis mixture-SPRT detector delivers a high-precision, long-lead alarm—a false-positive rate of 0.048 (lowest by $\approx 4\times$), a strict win over a leverage tripwire at matched recall, and a median lead of 27.9 days—with the signal carried almost entirely by the exposure axis. The honest cost is coverage: recall is $\approx 24\%$, the cohort is scaled and indicative, and the subclass is for now defined post hoc. The contribution is a truthful map of where behavioral early warning works and where it cannot—a low-FPR overlay for the gradual-leverage subclass, paired with an equity stop for the sudden majority—rather than a flashy claim that the data do not support.

A Reproducibility

All inputs are public and unauthenticated. We use the Hyperliquid `info` endpoint [10] for fills, orders, ledger updates, and the `portfolio` account-value history, and the public leaderboard snapshot for the universe. The fetcher throttles to ≥ 0.4 s between calls with retry on HTTP-429, and raw JSON responses are archived so the cohort can be rebuilt without re-querying. The detector (per-trade primitives $\rightarrow$ bucket-close per-axis observations $\rightarrow$ per-axis mSPRT $\rightarrow$ Holm min-gate, sustained- M composite $\rightarrow$ 70%-peak / pre-first-liquidation guard band), the baselines B1–B3, the early-window baseline with stability guard, and the event-clustered bootstrap are released as code alongside this paper. The cohort definition (universe filter, T_0 ,

window, labeling rule, market-event tag) and the detector form are pre-specified; the operating point is chosen on the tuning split only (Appendix C).

B The forward-only blow-up rule

A master is labeled a blow-up at the earliest time t satisfying either: (i) a peak-to-trough drawdown from the running pre- t equity peak exceeding 70%, with no subsequent recovery to 80% of that peak within 30 days; or (ii) the first on-chain liquidation fill. The rule is evaluated forward only—it uses no information after t —so a label cannot leak into the detection window. For genuine-loss filtering on the equity path we additionally require the cumulative-PnL drop across the crater to be at least half the equity drop, removing withdrawals masquerading as losses. Robustness reruns vary the drawdown threshold over $\{0.50, 0.70, 0.85\}$.

C Operating-point selection procedure

The detector grid varies sustain $M \in \{2, 3, 4\}$ and shrinkage $\kappa \in \{7, 14\}$ at fixed $\tau = 0.3$, 4-hour buckets, and a 20-bucket burn-in. On the *tuning* split only (32 blow-up / 41 stable) we retain grid points with tuning stable-FPR ≤ 0.10 , select the highest-recall configuration, and break ties by the longest tuning lead. This yields the reported operating point $M = 2$, $\kappa = 7$. The held-out evaluation cohort (63 blow-up / 63 stable) is never used in this selection; all Section 5 numbers are computed on it. Under this operating point the held-out blow-up cohort splits into 40 with a clean early-window baseline and 23 falling back to the relaxed baseline, with none excluded outright—the relaxed stability guard converts the previously high exclusion rate into a reported fallback rather than dropped masters. As noted in Section 4, κ is effectively inert here because the self and universe distributions nearly coincide on-chain; it is retained for the future run in which they differ.

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